

Muhammad Atif

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SUMMARY

Senior Data & Cloud Architect with 12+ years of progressive experience designing, building, and modernizing enterprise data platforms across healthcare, financial services, SaaS, fintech, and e-commerce. Deep hands-on expertise across cloud-native data engineering, lakehouse architecture, batch and real-time processing, distributed systems, data modeling, integration, governance, security, and platform reliability using AWS, Azure, Databricks, Snowflake, Spark, Kafka, Python, SQL, and Kubernetes. Proven ability to define target-state and reference architectures, translate business and technical requirements into scalable solutions, and lead engineering teams from architecture and design through implementation and production operations. Experienced in building secure, governed data platforms that enable analytics, AI/ML, self-service data products, and business-critical workloads in regulated enterprise environments.

SKILLS

Data Engineering & Languages: Python (Pandas, Polars, SQLAlchemy), Advanced SQL (T-SQL, PL/pgSQL), PySpark, Apache Spark, Spark SQL, Scala, Java, Linux, Bash, PowerShell, Go, TypeScript, C#, Git

Data Architecture, Lakehouse, Warehousing & Modeling: Databricks (Workflows, Delta Live Tables/Lakeflow, Photon, Auto Loader, Z-Ordering, Liquid Clustering), Delta Lake, Snowflake, Amazon Redshift, Azure Synapse, Google BigQuery, Microsoft Fabric/OneLake, dbt, Medallion Architecture, Apache Iceberg, Apache Hudi, Kimball Dimensional Modeling, Inmon 3NF, Data Vault 2.0, SCD Type 1/2/3, Conceptual/Logical/Physical Data Modeling, Physical Database Design, Indexing, Partitioning, Query Optimization, Performance Tuning

Databases: PostgreSQL, MySQL, SQL Server, Oracle, MongoDB, Neptune, Neo4j, OpenSearch, pgvector, Weaviate, ChromaDB, Redis, Amazon Aurora, RDS, Hive

ETL/ELT & Orchestration: Apache Airflow (Astronomer), Hadoop, AWS DMS, Step Functions, Prefect, Dagster, AWS Glue, Azure Data Factory, Informatica PowerCenter, SSIS, Talend, Fivetran, Airbyte, Apache NiFi, Boomi, Pega, REST API Integration, FastAPI, Data Contracts, Schema Evolution, Incremental Processing, Backfills, Cloud Composer, Matillion, JSONB Modeling, XML Processing, Parquet, Avro

Streaming & Event-Driven Data: Apache Kafka, Confluent, Flink, Spark Structured Streaming, Debezium CDC, AWS Kinesis, Azure Event Hubs, Google Pub/Sub, Event-Driven Architecture, Stateful Stream Processing

Cloud Data Platforms: AWS (S3, EC2, ECS, Fargate, EMR, Glue, Lambda, Kinesis, SQS, SNS, EventBridge, DynamoDB, Redshift, Athena, Lake Formation, EKS, SageMaker, Aurora, RDS, CloudWatch); Azure (ADLS Gen2, Databricks, Synapse, Data Factory, Fabric, Purview, Azure DevOps, Azure Key Vault); GCP (Big Query, Dataflow, Apache Beam, Dataproc, Pub/Sub, GKE, Bigtable, Vertex AI, Spanner)

Data Governance, Quality & Security: Unity Catalog, Microsoft Purview, Collibra MDM, Alation, Great Expectations, Monte Carlo, Open Lineage, RBAC, Row-Level Security, Column Masking, Data Lineage, Data Quality, Data Contracts, DAMA-DMBOK, CDMP, HIPAA, HITRUST, SOX, GDPR, CPRA, SOC 2, HL7/FHIR, X12 EDI, CCD/C-CDA, ICD-10/CPT/NDC/RxNorm/NPI, VPC/Transit Gateway, IAM/RBAC (cloud-infra flavor), Microsoft Entra ID, Privileged Identity Management (PIM), KMS, Zero-Trust

DevOps, IaC, Observability & FinOps: Terraform, AWS CDK, CloudFormation, Bicep/ARM Templates, Docker, Kubernetes, Helm, Jenkins, GitHub Actions, GitLab CI, Argo CD, Azure DevOps, pytest, Prometheus, Grafana, Datadog, Open Telemetry, FinOps Cloud Cost Optimization

Architecture & Technical Leadership: TOGAF-Aligned Architecture Practices, C4 Model, ADRs, Erwin, Domain Driven Design, Data Mesh, Event-Driven Architecture, Reference Architecture, ERD/UML, TCO Modeling, SOW/RFP Development, Architecture Reviews, Executive Stakeholder Communication, Technical Mentoring, Agile/Scrum, SDLC, Recovery Planning

AI-Ready Data Platforms & Tooling: LangChain, LangGraph, Llama Index, RAG, Pinecone, FAISS, Hugging Face, ML flow, AWS Sage Maker, Vertex AI, Model Context Protocol (MCP), Snowflake Cortex, Bedrock, Open AI, Fuzzy Matching & Entity Resolution, PDF/Document Extraction, GitHub Copilot, Claude / Claude Code, Cursor

BI/Reporting Layer: Power BI, Tableau, Looker, Semantic Layer (dbt, LookML)

EXPERIENCE

Data Solution Architect

Omnicell

January 2023 - Present

- Led the **architecture and technical design of cloud-native healthcare data platforms** using **Databricks, Apache Spark, Python, SQL, and cloud data services**, resolving fragmented requirements across clinical, business, security, compliance, and engineering teams by translating them into **target-state/reference architectures, data models, integration patterns, NFRs, and implementation roadmaps** that guided production delivery.

- Architected a **Patient 360 lakehouse** to address fragmented patient information across **6+ healthcare domains**, integrating **Epic, Cerner, FHIR, X12 EDI, claims, medications, laboratory, patient, encounter, and provider data** into a governed longitudinal record that enabled **interoperability, enterprise analytics, AI/ML, and reusable data products** from a consistent patient-data foundation.
- Replaced one-size-fits-all data movement with **4 reusable enterprise integration patterns using batch, streaming, API, and CDC** selecting architectures based on **volume, latency, availability, recovery, security, and cost requirements** to support **near-real-time clinical events** while maintaining efficient historical and bulk processing.
- Addressed inconsistent patient, provider, and clinical definitions by establishing **canonical data models, data contracts, schema-evolution standards, and Master Data Management (MDM) patterns** across **6 core healthcare domains**, applying **EMPI/MRN identity resolution and reference-data standardization** to create consistent identities and definitions across **EHR, claims, laboratory, and analytical systems**.
- Designed a **HIPAA/HITRUST-aligned data governance and security architecture** to protect sensitive **PHI** across **9+ controls**, implementing **least-privilege access, patient-level security, RBAC, masking, encryption, lineage, auditability, and controlled data access** so healthcare data could be safely consumed across engineering, analytics, clinical, and AI workloads.
- Engineered reusable **data reliability and observability patterns** for **5 critical production failure scenarios** **schema drift, duplicate events, malformed HL7/FHIR messages, delayed feeds, and incomplete records** using **validation, quarantine, reconciliation, replay, monitoring, and recovery controls** to prevent unreliable data from propagating into trusted downstream applications.
- Extended the governed **Patient 360** foundation into **predictive-care and RAG/GenAI workflows**, combining governed clinical data, **MLflow-managed models, permission-aware retrieval, evidence grounding, and human-in-the-loop controls** to support **readmission-risk prediction and clinician-facing AI use cases** while preserving security, traceability, and responsible data access.
- Owned **end-to-end solution architecture** from **requirements discovery and solution evaluation through design, implementation, architecture reviews, and production rollout**, converting clinical and business requirements into **architecture decisions, data standards, security controls, integration patterns, non-functional requirements, and technical roadmaps** while aligning **engineering, analytics, security, compliance, and clinical stakeholders**.
- Provided technical direction and architectural leadership to **4–5 data engineers**, establishing **engineering standards, architecture guardrails, design/code review practices, and production-readiness expectations** while mentoring engineers in **solution design, Spark/SQL performance, data modeling, troubleshooting, and technical decision-making**, improving consistency and reuse across healthcare data initiatives.

Senior Data Engineer

NICE

May 2020 – December 2022

- Modernized fragmented, batch-dependent financial data processing by building **high-volume batch and real-time pipelines** across **5 financial domains** **card payments, ACH/wire transfers, customer accounts, merchant transactions, and third-party risk feeds** using **Python, PySpark, SQL, Apache Kafka, Spark, and Databricks**, enabling **near-real-time data availability** for downstream fraud, risk, finance, and analytical workloads.
- Engineered resilient **Spark Structured Streaming** pipelines to address **duplicate, late-arriving, out-of-order, and replayed financial events**, applying **event-time processing, watermarking, deduplication, checkpointing, and stateful transformations** to maintain reliable transaction processing and provide **trusted event streams for fraud and risk analytics**.
- Re-engineered costly full-load processing into **Bronze, Silver, and Gold Delta Lake pipelines** across customer, account, transaction, merchant, counterparty, and payment domains, using **CDC, incremental processing, schema evolution, Delta MERGE, and idempotent processing patterns** to process changing financial data without repeatedly rebuilding large historical datasets.
- Prevented inaccurate transaction data from propagating into business-critical reporting by implementing automated **data-quality, reconciliation, and exception-management controls** across **5 validation areas** like **transaction counts, monetary totals, duplicates, rejected records, and schema changes** using **validation rules, quarantine patterns, and reconciliation checks** before data reached **finance, risk, regulatory, and analytical consumers**.
- Solved historical traceability and reporting inconsistencies by designing **Data Vault 2.0, SCD Type 2, and Kimball dimensional models** across customer, account, transaction, and payment domains, preserving historical changes and delivering **auditable, analytics-ready datasets** for **financial reconciliation, regulatory reporting, Power BI, and Tableau**.
- Addressed performance and compute inefficiencies in large **Databricks and Snowflake** workloads through **incremental processing, Photon, Liquid Clustering/Z-Ordering, query profiling, partition/data-layout optimization, and workload tuning**, improving the efficiency and scalability of **high-volume financial transformations** while reducing unnecessary compute consumption.
- Established **DataOps and production-reliability practices** across the end-to-end pipeline lifecycle using **Apache Airflow, automated data-quality validation, CI/CD, observability, and production monitoring**, tracking **data freshness, Kafka consumer lag, schema drift, processing latency, and pipeline failures** while performing **root-cause analysis, controlled backfills, event replays, and incident recovery** to restore trusted data processing without unnecessary full reloads.
- Led technical delivery across complex **data-engineering and platform workstreams**, establishing consistent **engineering standards, code/design reviews, automated testing, deployment practices, and production-readiness controls** while mentoring engineers in **PySpark/SQL performance tuning, streaming architecture, dimensional modeling, pipeline design, and production troubleshooting**, improving engineering consistency and maintainability across the data platform.

Cloud Engineer

Banner Health

January 2017 - April 2020

- Modernized a **manually managed B2B SaaS environment** into a **containerized AWS platform on Amazon EKS**, decomposing workloads into independently deployable services across **QA, staging, and production** to improve **scalability, deployment isolation, operational consistency, and failure containment** as application workloads grew.
- Eliminated repetitive and error-prone infrastructure setup by developing reusable **Infrastructure as Code (IaC)** modules with **Terraform** across **6 core infrastructure domains** that include **networking, EKS, IAM, KMS, databases, and load balancing**, creating standardized and repeatable provisioning patterns across multiple environments.
- Addressed variable workload demand by implementing **Kubernetes orchestration and autoscaling patterns** using **Docker, Amazon EKS, Helm, Argo CD, Horizontal Pod Autoscaler (HPA), and cluster autoscaling**, allowing customer-facing services and background workloads to scale independently during traffic changes while improving resource utilization and controlling unnecessary compute consumption.
- Replaced manual and inconsistent release processes with an automated **CI/CD delivery pipeline** spanning 5 stages **testing, security scanning, Terraform validation, approval, and deployment** using **GitHub Actions and Azure DevOps**, implementing **rolling/canary deployment strategies, production health checks, and automated rollback** to reduce deployment risk and improve release consistency.
- Strengthened security for **multi-tenant SaaS workloads** across **5 layers APIs, application services, databases, event flows, and cloud infrastructure by implementing least-privilege IAM, workload identities, secrets management, KMS encryption, network isolation, and private connectivity**, reducing unnecessary exposure of sensitive workloads and embedding **security-by-design** into the platform architecture.
- Standardized inconsistent cloud-environment creation by designing reusable **AWS and Azure landing-zone blueprints** covering **networking, identity and access management, security, logging, monitoring, governance, and CI/CD**, establishing repeatable cloud foundations for provisioning and operating enterprise workloads across environments.
- Defined and implemented **non-functional requirements (NFRs)** for **scalability, high availability, resilience, security, observability, recoverability, and cost efficiency**, strengthening production operations through **Multi-AZ architecture, autoscaling, recovery testing, resource rightsizing, monitoring, and non-production scheduling** while balancing **performance, operational resilience, and cloud cost optimization**.
- Provided hands-on technical guidance to junior engineers through **Terraform and Kubernetes reviews, cloud-infrastructure design, deployment troubleshooting, production incident analysis, and recovery exercises**, promoting reusable **IaC, DevOps, security, and operational-readiness standards** across cloud engineering initiatives.

ETL & Data Warehouse Engineer

PBG

January 2014 - December 2016

- Consolidated fragmented operational data from **6+ source types**, including transactional databases, payment systems, merchant APIs, settlement files, CRM, and order management systems, by building production **ETL/ELT pipelines using Python, SQL, database native processing, and enterprise ETL patterns**, creating centralized analytical datasets for **finance, operations, and business reporting**.
- Reengineered inefficient full load processing into **incremental and Change Data Capture (CDC) pipelines** across **4 core domains**, including transactions, customers, orders, and inventory, implementing **change based processing, restartable workflows, safe reruns, backfills, and late arriving data handling** to reduce unnecessary reprocessing and improve the reliability of scheduled data loads.
- Resolved inconsistent reporting definitions by designing **Kimball dimensional models** across **8+ business entities**, including transactions, settlements, customers, merchants, orders, products, payments, and inventory, defining **fact table grain, surrogate keys, relationships, and conformed dimensions** to provide consistent and reusable data structures for **financial and operational reporting**.
- Preserved historical business context by implementing **Slowly Changing Dimension (SCD) Type 2** processing for **customer, merchant, and product dimensions**, maintaining effective dated historical attributes so reports and analytical workloads could accurately reconstruct the business state associated with prior **transactions, orders, and settlements**.
- Addressed financial data discrepancies by building automated **reconciliation, data quality, and exception management controls** across **4 critical checkpoints**, including payment processors, internal transaction systems, bank settlements, and the general ledger, validating **record counts, monetary totals, duplicates, and unmatched transactions** while routing exceptions for investigation before data reached trusted financial reporting.
- Improved production data reliability by orchestrating **end to end ETL workflows across 6 processing stages**, covering source validation, ingestion, transformation, data quality validation, reconciliation, and data mart refresh, while performing **root cause analysis, failed job recovery, SQL performance tuning, and incremental workload optimization** to maintain dependable delivery of analytics ready datasets to downstream **BI and reporting consumers**.

PROJECT

Real-Time Enterprise CDC & Event-Driven Integration Platform

- Architected a reusable event-driven CDC platform across 4 source categories including relational databases, APIs, SaaS applications, and legacy systems, using Debezium, Kafka Connect, Apache Kafka, Flink, and Spark Structured Streaming to replace batch-dependent integrations with near-real-time data delivery.

- Implemented 7 resilient streaming controls covering schema evolution, idempotent processing, event-time handling, checkpointing, replay, dead-letter queues, and automated reconciliation, improving recovery from late, duplicate, malformed, and failed events without requiring full pipeline reloads.
- Established governed delivery across Delta Lake and Snowflake consumption layers, implementing end-to-end observability with OpenTelemetry, Prometheus, and Grafana to monitor 5 critical production signals including consumer lag, processing latency, data freshness, failed events, and schema drift.
- Developed reusable CDC and streaming patterns supporting 3 downstream workload types including operational analytics, enterprise applications, and governed data products, improving data freshness while decoupling source systems from downstream consumers and strengthening production recovery.

Enterprise Data Mesh & Self-Service Data Product Platform

- Architected an enterprise Data Mesh and self-service platform across multiple business domains, establishing standardized ownership, governance, security, quality, and engineering guardrails for independently managed data products.
- Designed reusable data-product patterns across 3 core platform layers using Databricks, Snowflake, dbt, and Airflow, standardizing data contracts, quality SLAs, schema evolution, and Bronze, Silver, and Gold publishing workflows.
- Established 3 layers of governed data management covering metadata management, enterprise cataloging, policy-based access, and end-to-end lineage using Unity Catalog, Collibra, and OpenLineage, exposing certified data products through governed semantic and consumption layers to improve data discovery, trusted consumption, and self-service analytics.
- Enabled domain teams to independently publish and consume governed data products through reusable semantic and consumption layers, reducing dependency on centralized engineering and accelerating self-service analytics delivery.

AI-Ready Governed Enterprise Data Access Platform

- Designed a governed AI-ready data access platform supporting 3 data domains: structured warehouse data, enterprise documents, and metadata, enabling AI assistants to retrieve business context without bypassing existing security and governance controls.
- Built permission-aware RAG workflows integrating MCP, LangChain/LangGraph, Snowflake Cortex, and vector search, providing unified retrieval across structured and unstructured enterprise data while preserving user-level access permissions.
- Implemented an end-to-end retrieval pipeline covering 5 core stages: document chunking, embedding generation, metadata enrichment, hybrid search, and grounding, improving the relevance and traceability of context supplied to downstream LLM applications.
- Enforced 4 layers of data protection: Unity Catalog governance, RBAC, row-level security, and sensitive-data controls, preventing unauthorized enterprise information from entering LLM context while maintaining lineage and auditability.
- Developed reusable AI data-access patterns combining permission-aware retrieval, evidence citations, grounding, and human-review controls, allowing multiple AI applications to securely consume governed enterprise data without maintaining separate point-to-point retrieval integrations.

CERTIFICATIONS

Databricks Certified Data Engineer Associate

Microsoft Fabric Data Engineer Associate

AWS Certified Data Engineer Associate

AWS Well-Architected Framework

Snowflake SnowPro Certification

EDUCATION

Bachelor of Science in Computer Science

Punjab University • 2014
